

Handcrafted Feature Detection and Extraction

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DaSCI

Instituto Andaluz de Investigación en
Data Science and Computational Intelligence

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- Corner Harris Detector
- Scale-Invariant Feature Transform (SIFT)
- Histogram of Oriented Gradients (HOG)

Readings

- ***Harris corner detector***
 - Forsyth and Ponce (2012), Chapter 5.3.1
 - Szeliski (2022), Chapter 7.1
 - González and Woods (2018), Chapter 11.6
- ***Scale-Invariant Feature Transform (SIFT)***
 - Forsyth and Ponce (2012), Chapter 5.4.1
 - Szeliski (2022), Chapter 7.1
 - González and Woods (2018), Chapter 11.7
 - Otero & Delbracio (2014), “Anatomy of the SIFT Method”
- ***Histogram of Oriented Gradients (HOG)***
 - Forsyth and Ponce (2012), Chapter 5.4.2
 - Szeliski (2022), Chapter 6.3.2

Corner Detection

Corner Detection

- See information in practical assignment 3.

Scale-Invariant Feature Transform (SIFT)

SIFT (Scale Invariant Feature Transform)

- Lowe, D. G. (2004). Distinctive image features from scale-invariant keypoints. *International journal of computer vision*, 60, 91-110. (>81.000 citations)
- Highly recommended reading:
 - "Anatomy of the SIFT Method" (Ives Rey-Otero, Mauricio Delbracio) Image Processing On Line, 2014 <http://dx.doi.org/10.5201/ipol.2014.82>
From Otero's PhD dissertation (École Normale Supérieure de Cachan-ENS Cachan): <https://theses.hal.science/tel-01226489/document>

SIFT (Scale Invariant Feature Transform)

- SIFT is a milestone in the combination of science and engineering within the field of computer vision
- SIFT is the result of applying/combining our knowledge on:
 - Gradient Magnitude and Orientation
 - Gaussian Pyramids
 - Laplacian of Gaussian approximation via Difference of Gaussians
 - Blob Detection
 - Automatic Scale Selection
 - Keypoint Detection and Description
 - Histogram Representation
 - ...

SIFT (Scale Invariant Feature Transform)

1) SIFT DETECTOR

- Detect interest points at different scales

2) ORIENTATION ASSIGNMENT

- Assign principal orientation to each interest point

3) SIFT DESCRIPTOR

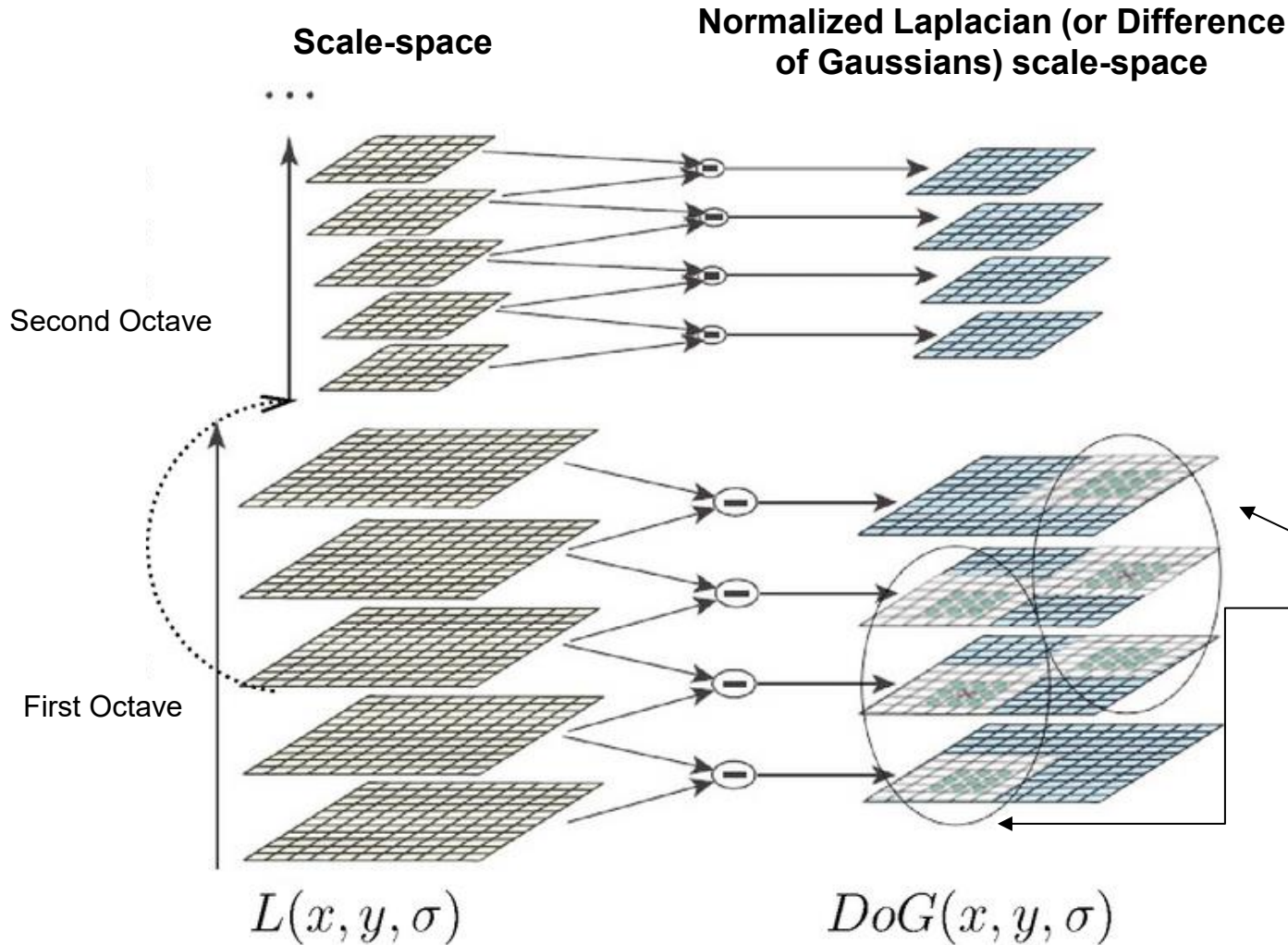
- Compute a vector for each keypoint such that the descriptor is highly distinctive

SIFT detector

Overall, similar to multiscale blob detection.

Detect the 3D discrete maxima and minima (the number of **neighbors** is $26=3\times3\times3-1$).

This scanning process is **sensitive to noise** and produces **unstable detections** → This step is followed by an **interpolation**, that refines the localization of the extrema, and by filtering to **discard unreliable detections** (edges vs corners, low-contrast regions).



Octave: sets of images sharing a common sampling rate.

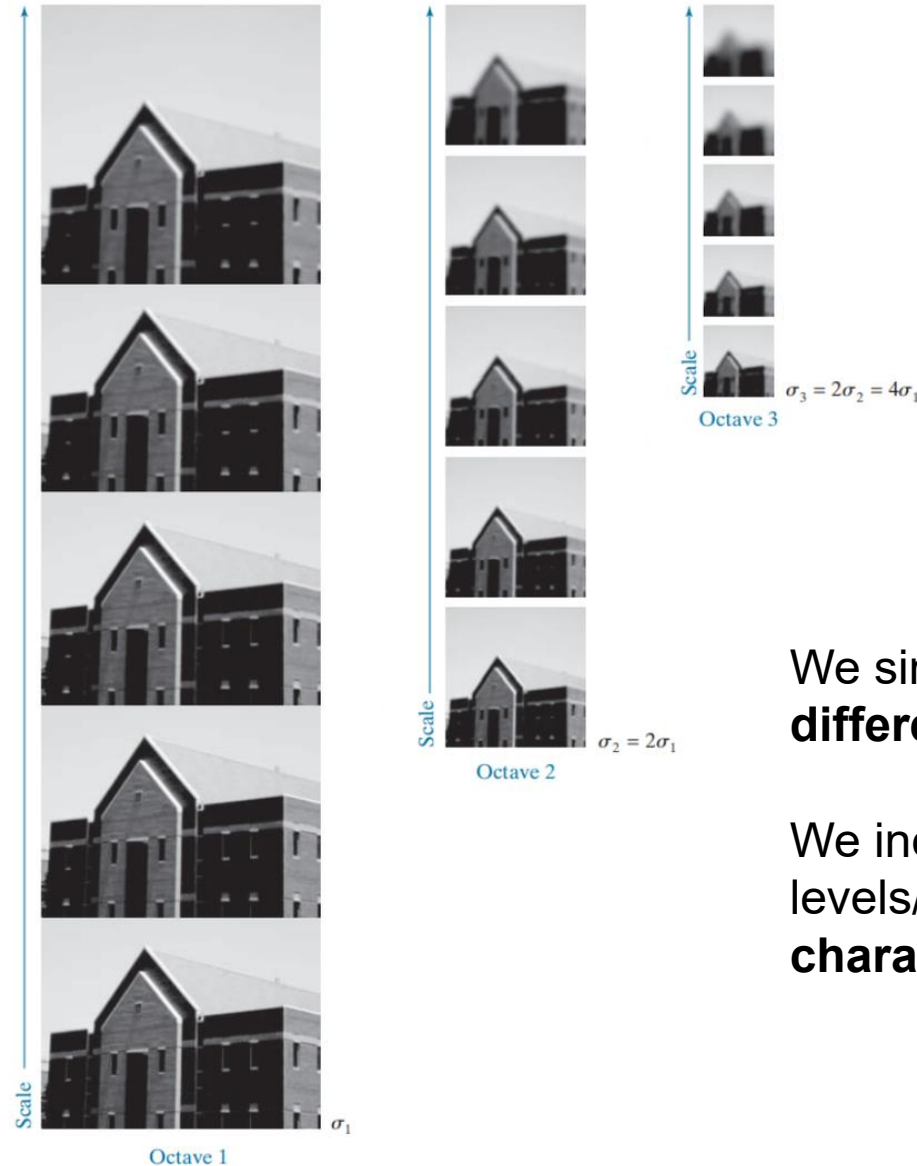
Each octave corresponds to a doubling of σ , just as an octave in music theory corresponds to doubling the frequency of a sound signal (González & Woods, 2018).

Many technical details related with the Gaussian scale-space and the keypoint definition are included in ["Anatomy of the SIFT Method"](#)

SIFT detector

Octave: sets of images sharing a common sampling rate.

Each octave corresponds to a doubling of σ , just as an octave in music theory corresponds to doubling the frequency of a sound signal (González & Woods, 2018).

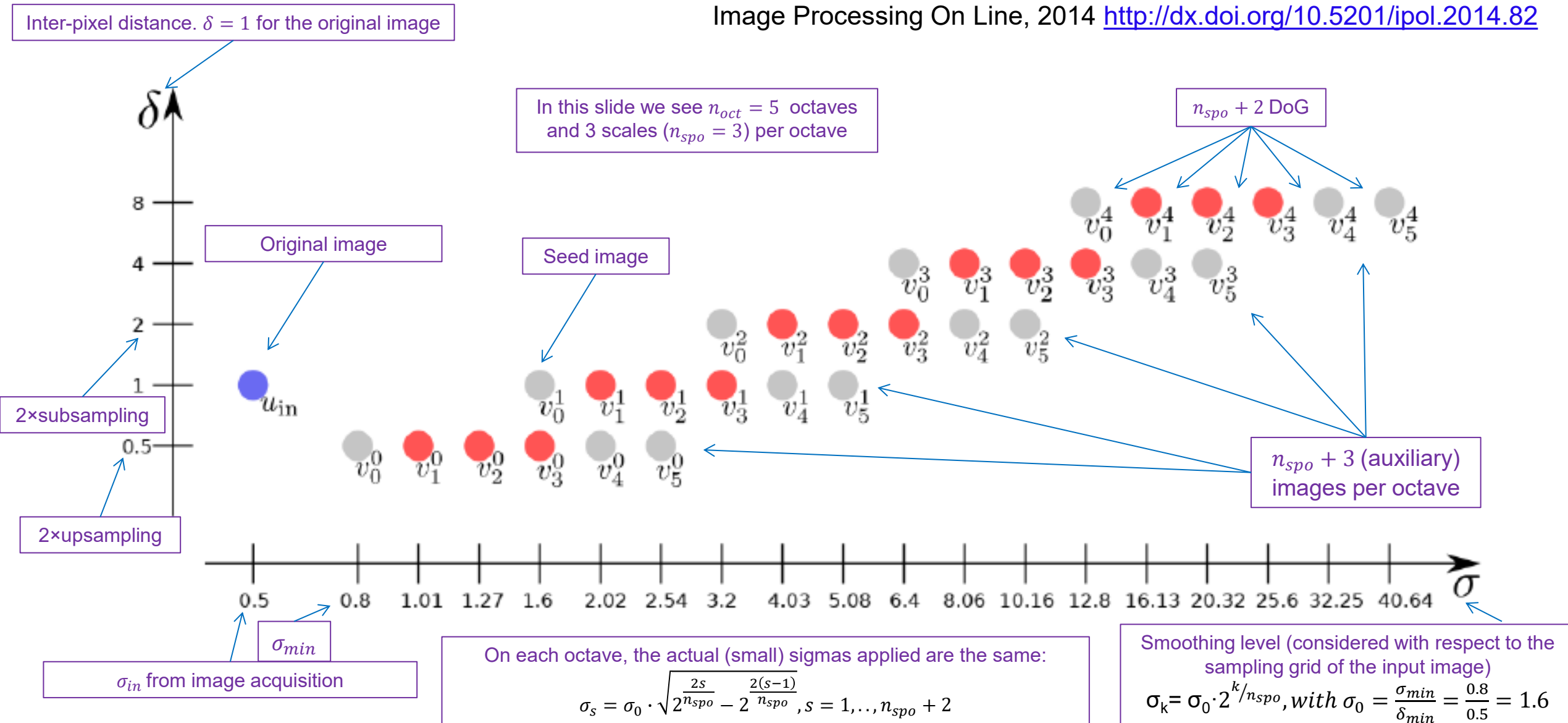


We simulate “**looking at the scene with different resolutions / distances**”.

We incorporate several intermediate levels/scales for a **finer detection of the characteristic scale of the blob/region**.

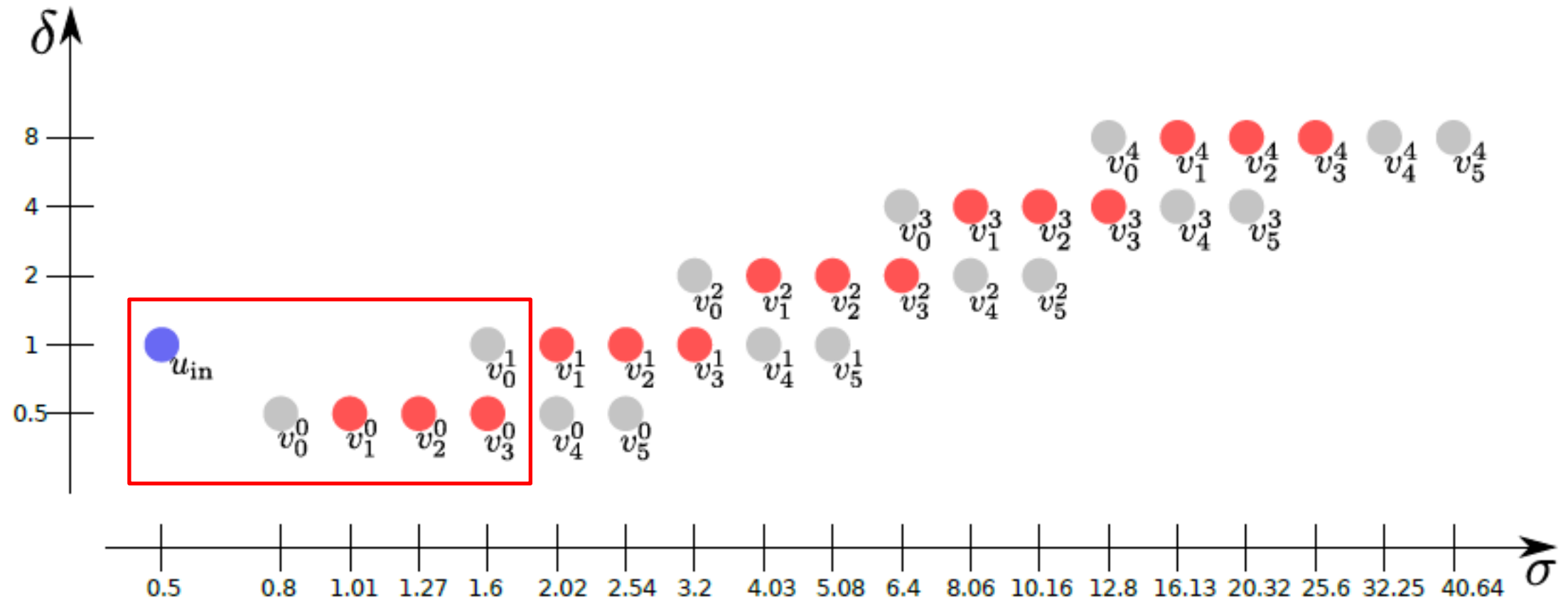
SIFT detector: Scale-space default configuration

"Anatomy of the SIFT Method" (Ives Rey-Otero, Mauricio Delbracio)
Image Processing On Line, 2014 <http://dx.doi.org/10.5201/ipol.2014.82>



SIFT detector: Scale-space default configuration

- Just as an example, we can focus on the construction of (part of) the first octave:



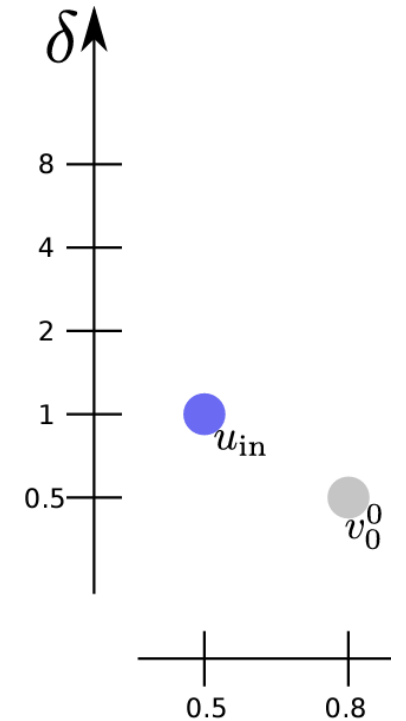
SIFT detector: Scale-space default configuration

- First, we need to go from $\sigma_{ini} = 0.5$ to $\sigma_{fin} = 0.8$. How?
 - We must smooth the image with $\sigma_{ini} = 0.5$ with «what's missing»

$$\sigma_{fin}^2 = \sigma_{ini}^2 + \sigma_{incr}^2$$

$$0.8^2 = 0.5^2 + \sigma_{incr}^2$$

$$\sigma_{incr} = \sqrt{0.8^2 - 0.5^2} = 0.624$$



Remember that

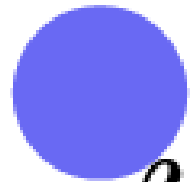
$$G(x, y | \sigma) = G\left(x, y \mid \sqrt{\sigma_1^2 + \sigma_2^2}\right) = G_1(x, y | \sigma_1) * G_2(x, y | \sigma_2)$$

Convolving twice with Gaussian kernel of width σ is the same as convolving once with kernel of width $\sqrt{\sigma^2 + \sigma^2} = \sqrt{2}\sigma = \sigma\sqrt{2} \rightarrow \sqrt{0.5^2 + 0.624^2} = 0.7996$

SIFT detector: Scale-space default configuration

Initial sigma (assumed blur level of the input image)

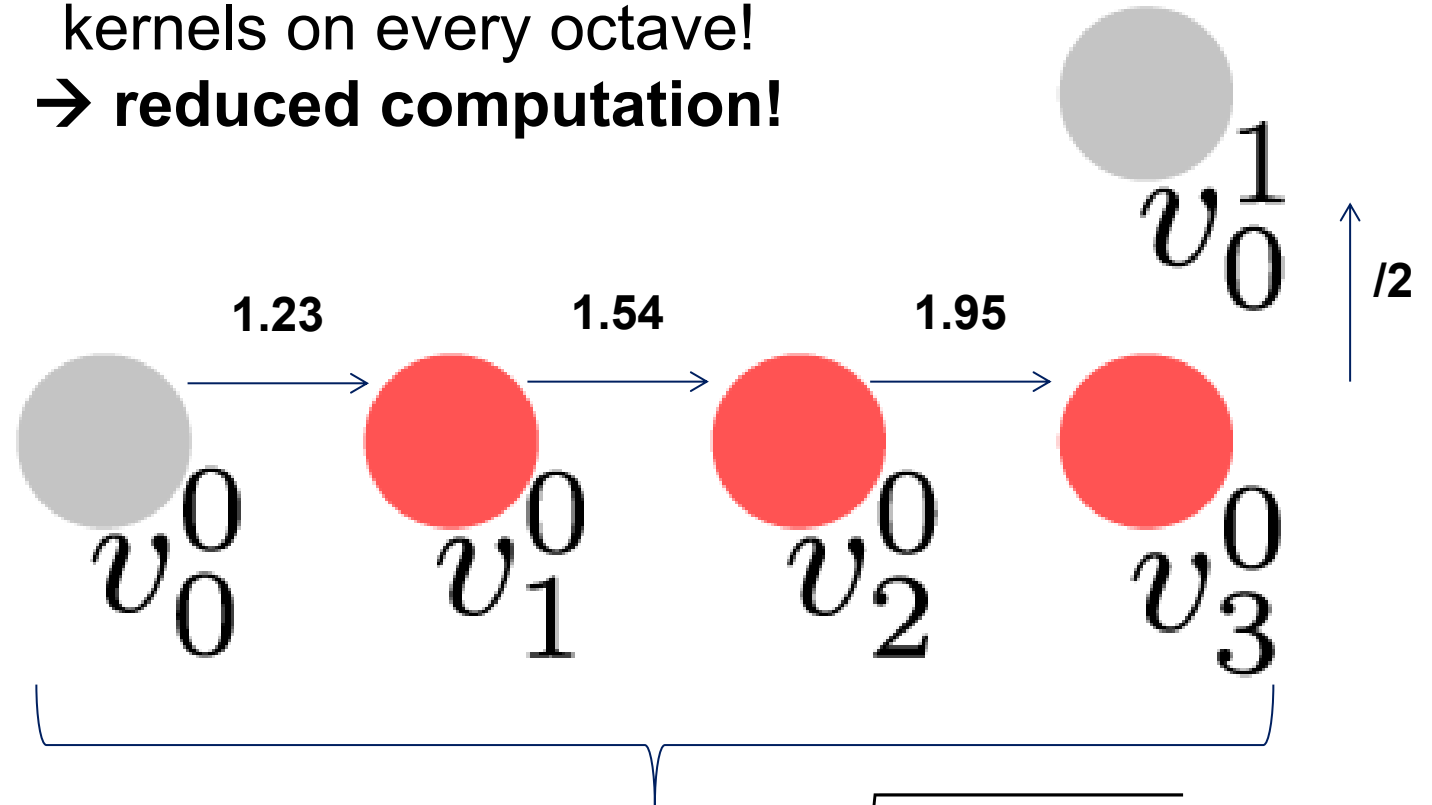
$$\sigma_{in} = 0.5$$



u_{in}

x2

SIFT proposes to smooth using always the same “small” Gaussian kernels on every octave!
→ **reduced computation!**



Sigmas calculated from

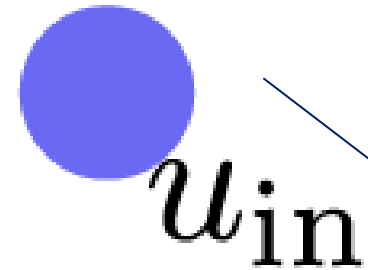
($\sigma_0 = 1.6$, $s = 1..5$, $n_{spo} = 3$)

$$\sigma_s = \sigma_0 \cdot \sqrt{2^{\frac{2s}{n_{spo}}} - 2^{\frac{2(s-1)}{n_{spo}}}}, s = 1, \dots, n_{spo} + 2$$

SIFT detector: Scale-space default configuration

Initial sigma (assumed blur level of the input image)

$$\sigma_{in} = 0.5$$



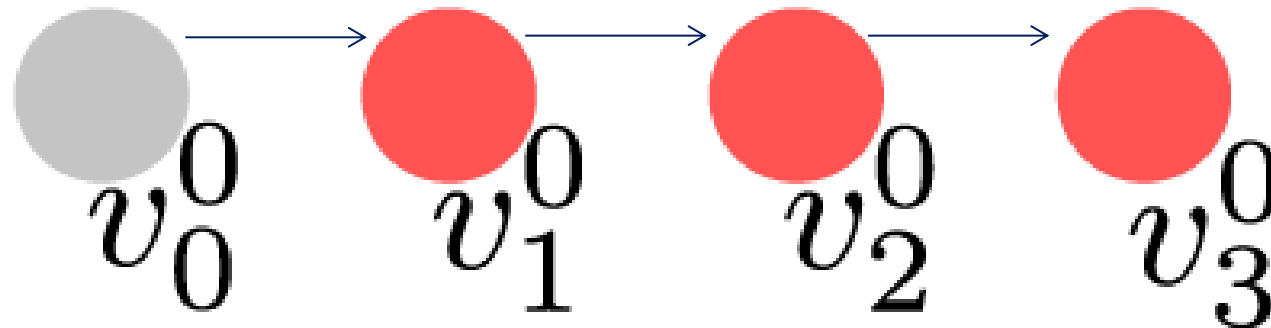
x2

We double the size of the image:

- Pixel size is halved per axis.
- **Same blur occupies twice the number of pixels.**

We halved the size of the image:

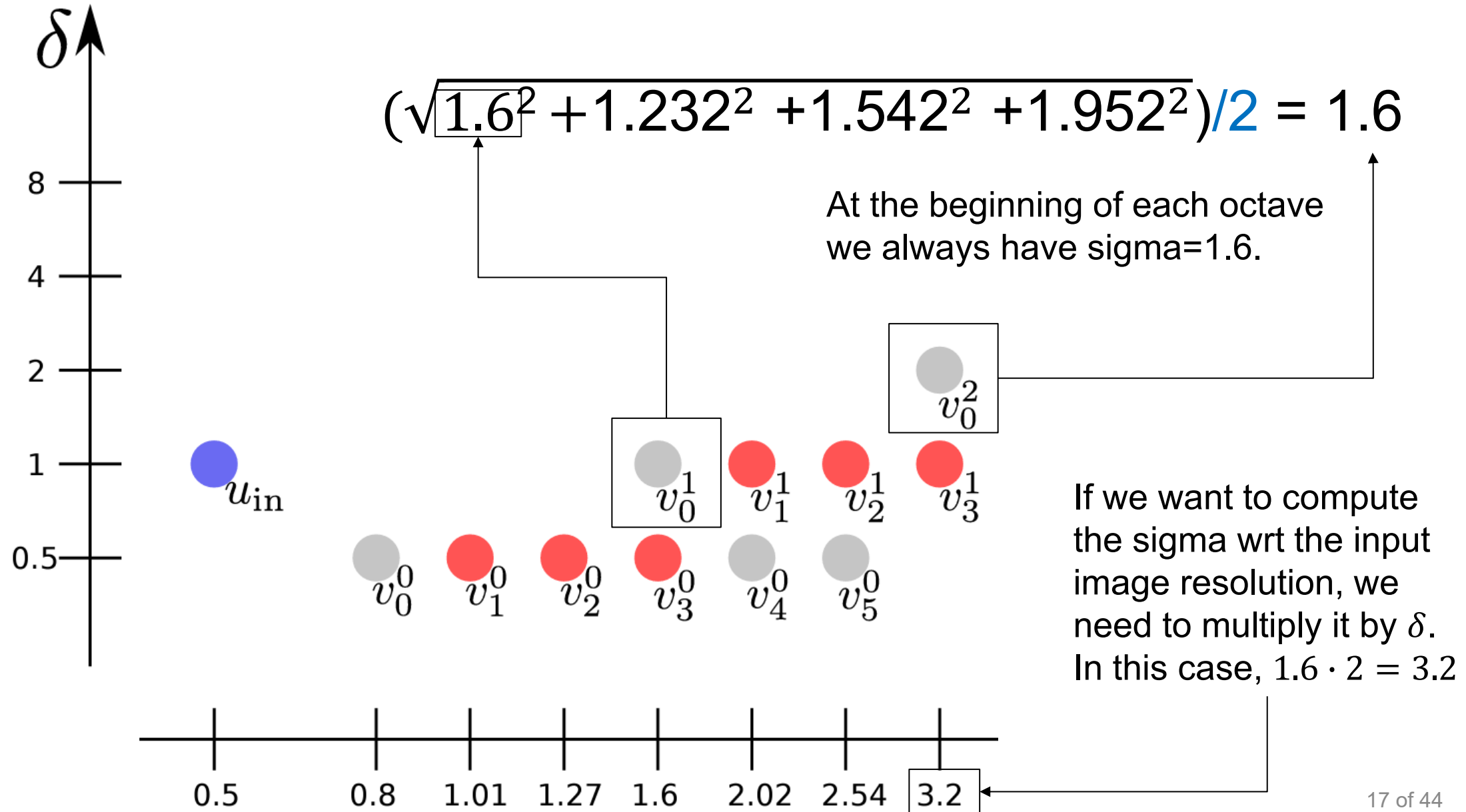
- Pixel size is doubled per axis.
- **Same blur occupies half the number of pixels.**



"I have blurred enough. Now I can keep one out of every two pixels."

$$(\sqrt{(\sqrt{0.5^2 + 0.624^2} \cdot 2)^2 + 1.23^2 + 1.54^2 + 1.95^2})/2 = 1.6$$

SIFT detector: Scale-space default configuration



SIFT detector: The DoG scale-space

"Anatomy of the SIFT Method" (Ives Rey-Otero, Mauricio Delbracio)
Image Processing On Line, 2014 <http://dx.doi.org/10.5201/ipol.2014.82>

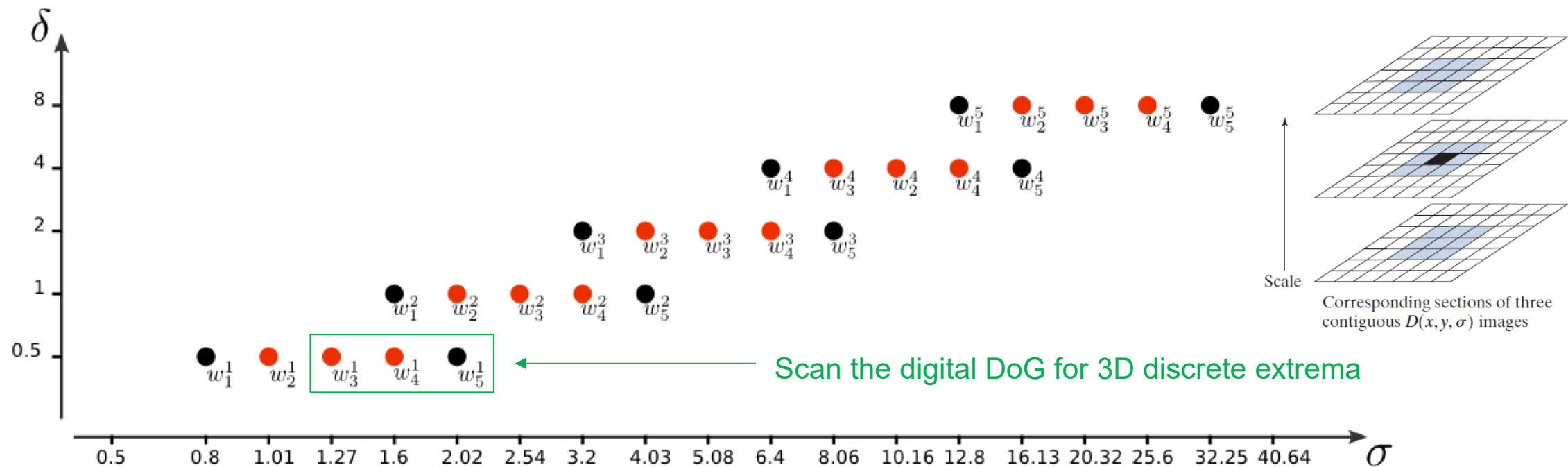
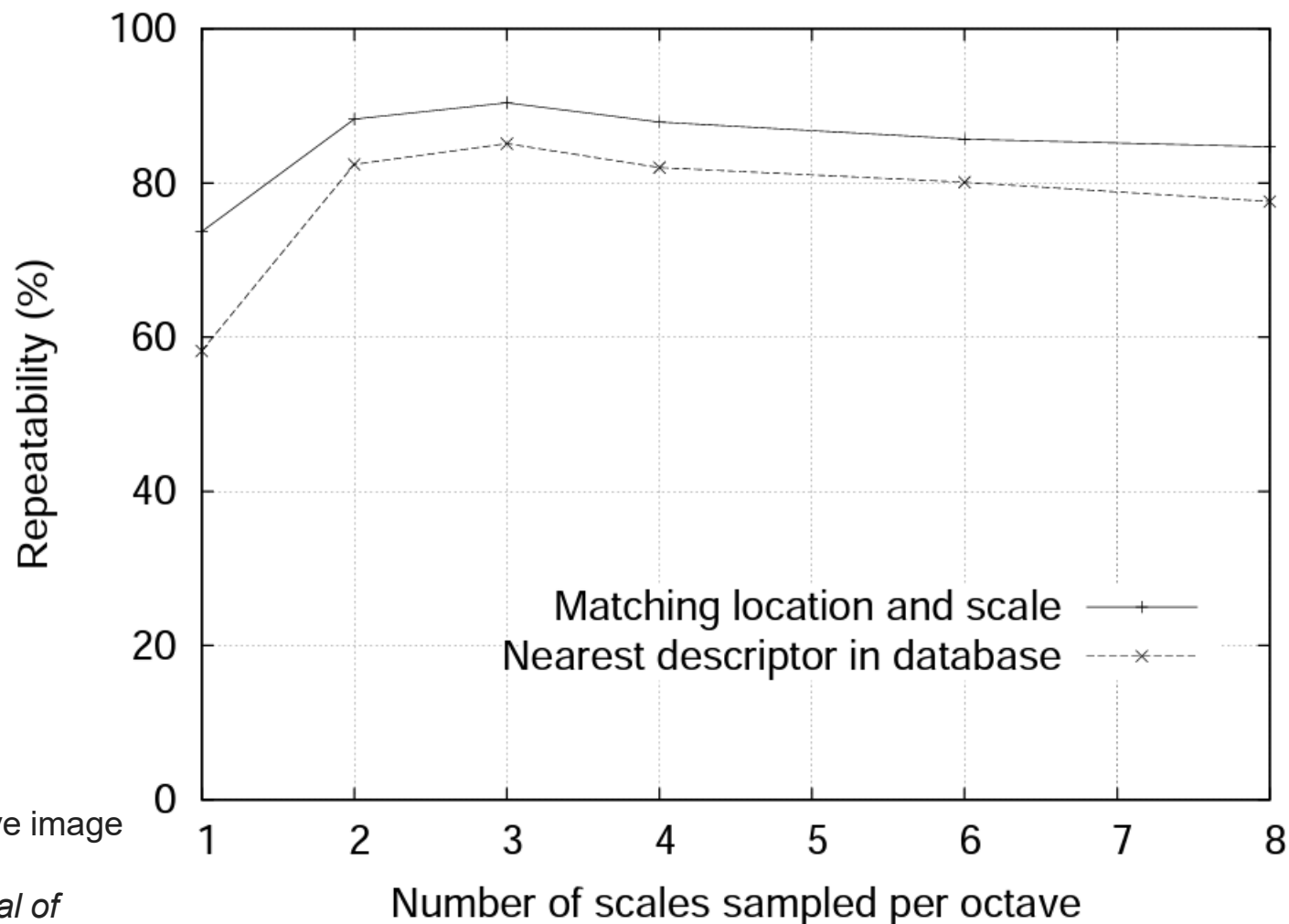


Figure 5: The *DoG* scale-space. The difference of Gaussians acts as an approximation of the normalized Laplacian $\sigma^2 \Delta$. The difference $\mathbf{w}_s^o = \mathbf{v}_{s+1}^o - \mathbf{v}_s^o$ is relative to the blur level σ_s^o . Each octave contains n_{spo} images plus two auxiliary images (in black).

SIFT detector: Scale-space default configuration



SIFT is **strongly heuristic**: it uses a significant number of **experimentally determined parameters**.

SIFT (Scale Invariant Feature Transform)

1) SIFT DETECTOR

- Detect interest points at different scales

The DoG/LoG is involved only in this stage!

2) ORIENTATION ASSIGNMENT

- Assign principal orientation to each interest point

3) SIFT DESCRIPTOR

- Compute a vector for each keypoint such that the descriptor is highly distinctive

We compute all this in Gaussian Scale-Space!

SIFT orientation assignment

- Take a square window around detected feature (in the corresponding scale σ_k)
- Compute gradient orientation (for each pixel) in the scale-space pyramid
- Create a weighted direction histogram (36 bins: 10° per bin). Weights are:
 - Gradient magnitudes

Arrows illustrate **gradient orientation** (arrow direction) and **gradient magnitude** (arrow length)

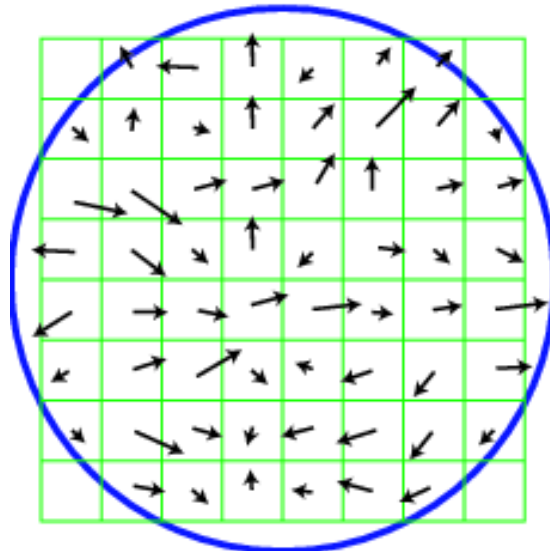
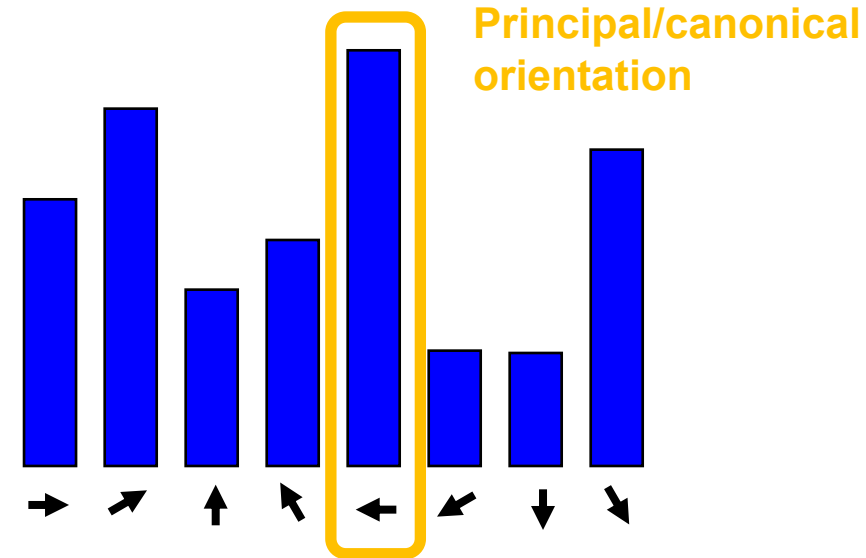


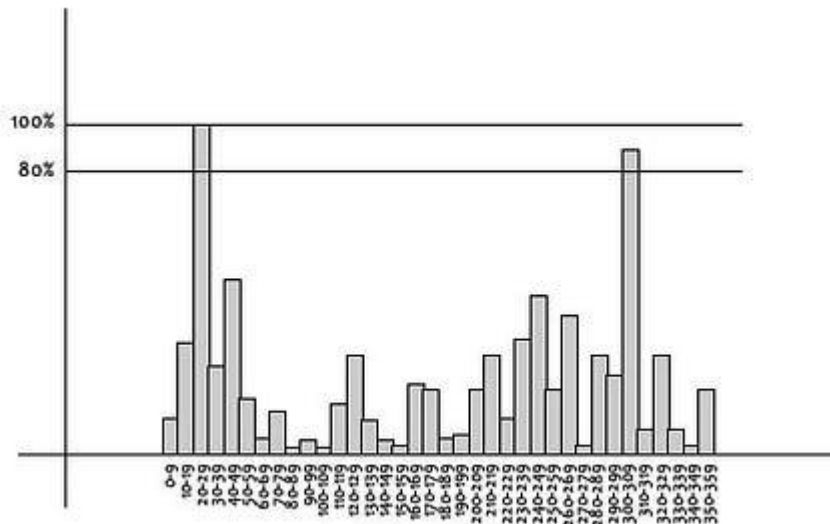
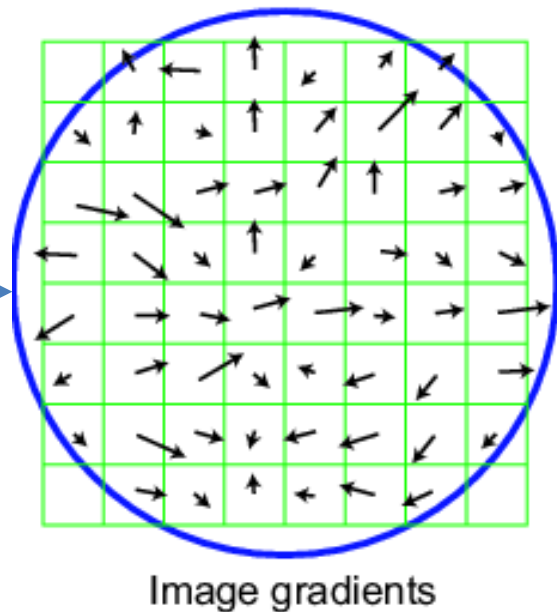
Image gradients



SIFT orientation assignment

- Take a square window around detected feature (in the corresponding scale σ_k)
- Compute gradient orientation (for each pixel) in the scale-space pyramid
- Create a weighted direction histogram (36 bins: 10° per bin). Weights are:
 - Gradient magnitudes
 - Spatial Gaussian filter with $\sigma = 1.5 \cdot \langle \text{keypoint_scale} \rangle$
(central pixels, close to detected keypoint, are more important)

Gradient magnitude is weighted
by a Gaussian window

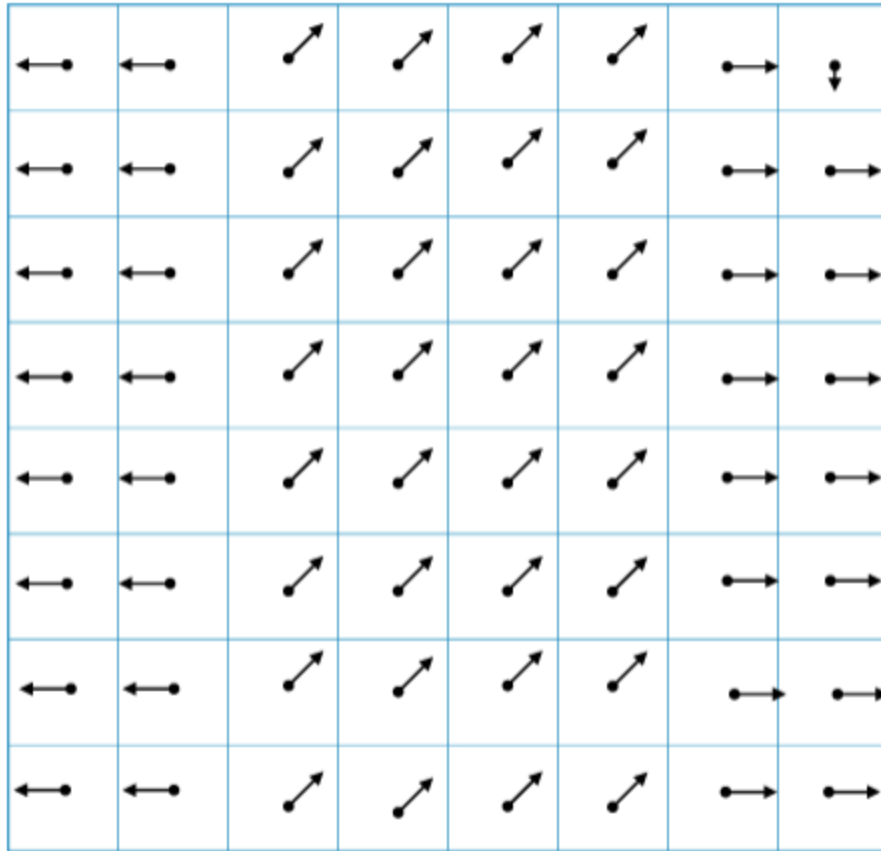


Peaks in the orientation histogram correspond to **dominant/reference directions** of local gradients.

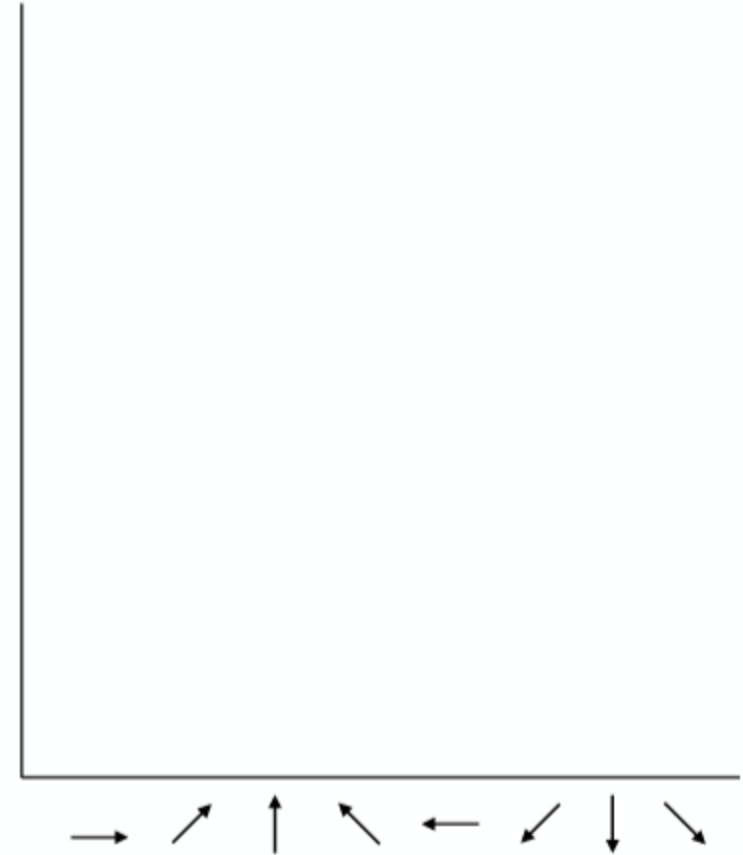
The highest peak is detected, and also any other local peak within 80% of the highest peak is used to create a keypoint with that orientation.

Each keypoint: **(x, y, scale, orientation)**

SIFT orientation assignment

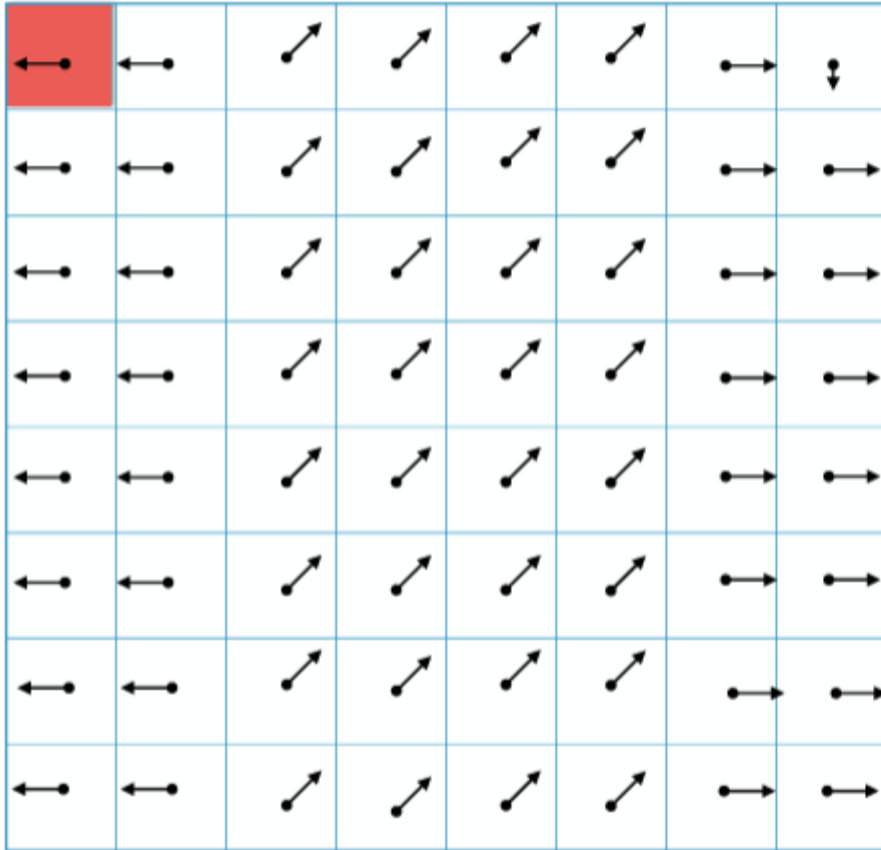


Arrows illustrate **gradient orientation** (direction)
and **gradient magnitude** (arrow length)

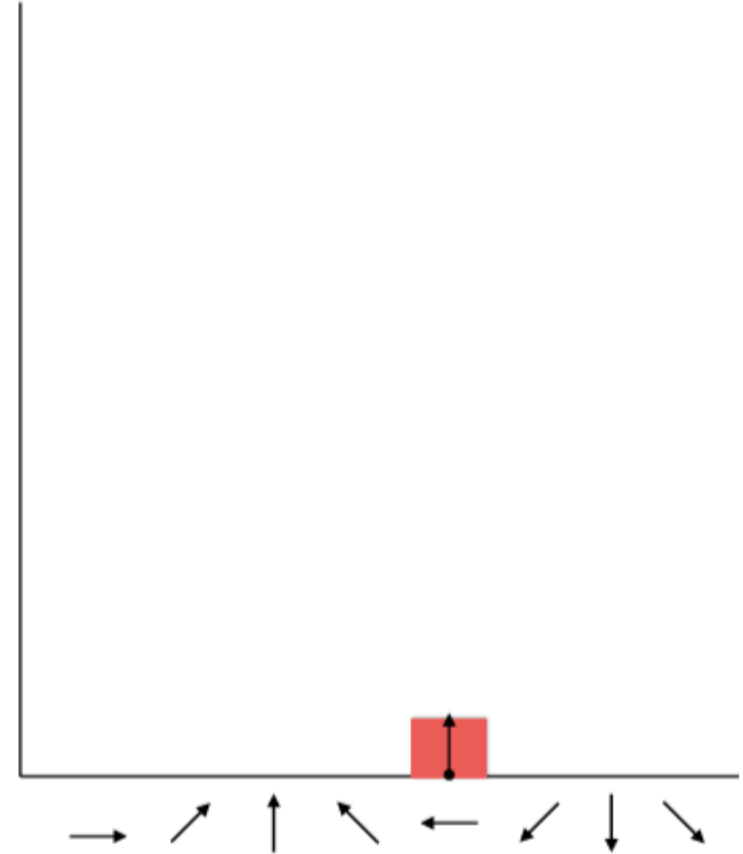


This visual example is taken from [Leonid Sigal slides](#) (The University of British Columbia)

SIFT orientation assignment

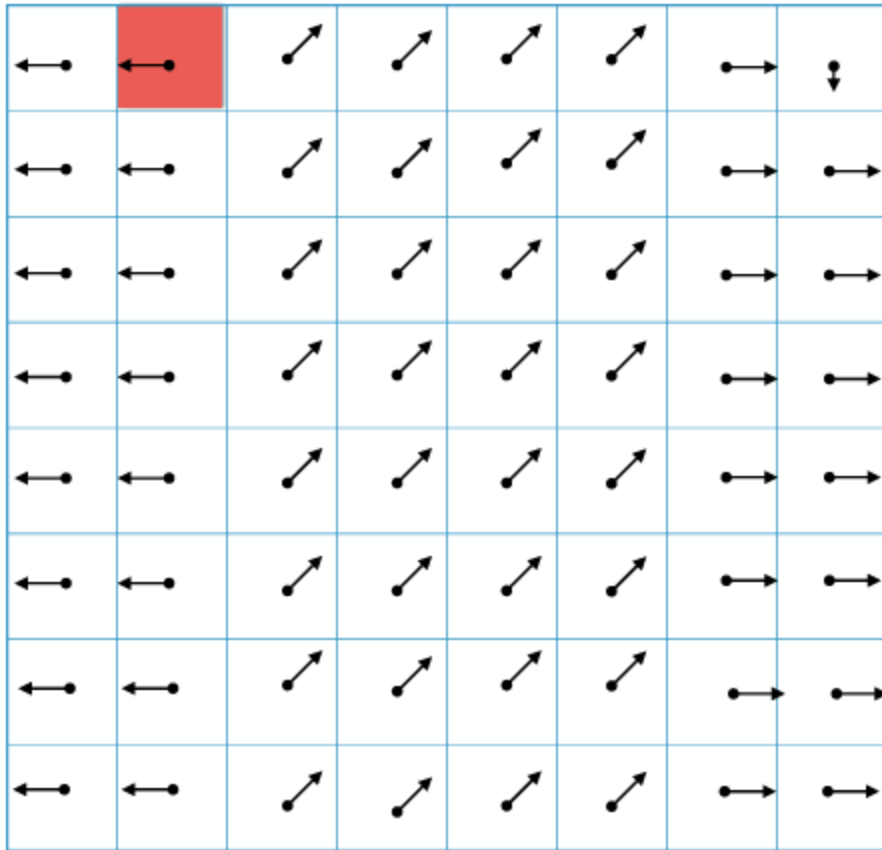


Arrows illustrate **gradient orientation** (direction)
and **gradient magnitude** (arrow length)

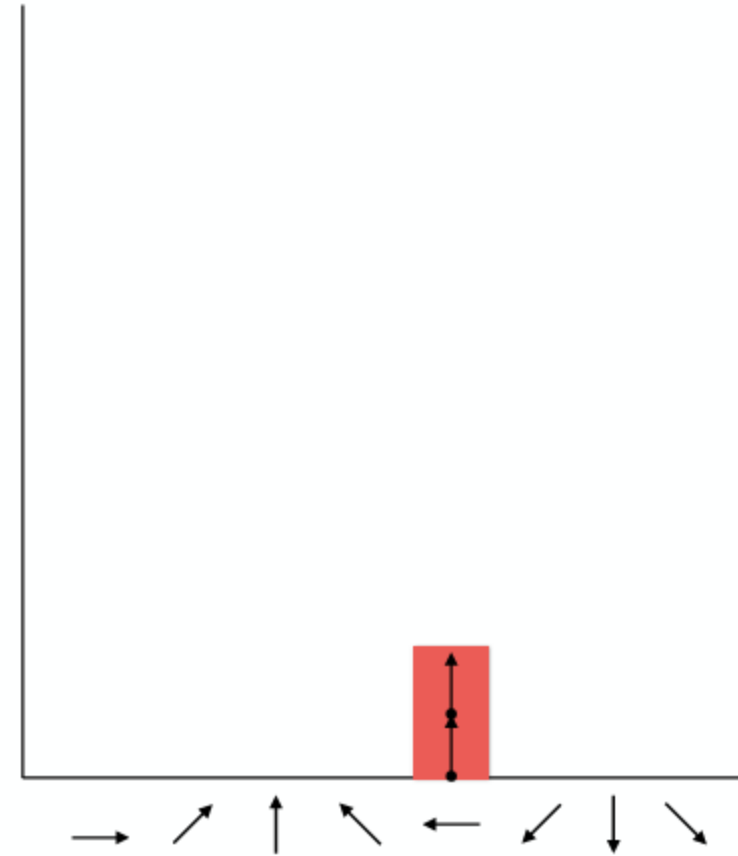


This visual example is taken from [Leonid Sigal slides](#) (The University of British Columbia)

SIFT orientation assignment

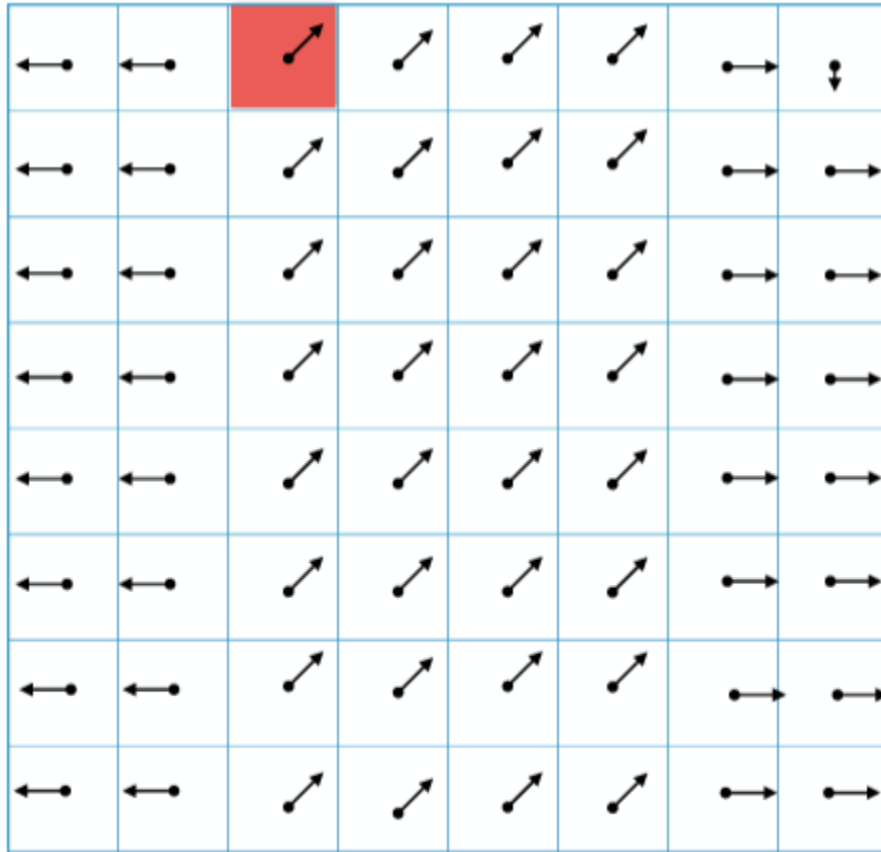


Arrows illustrate **gradient orientation** (direction)
and **gradient magnitude** (arrow length)

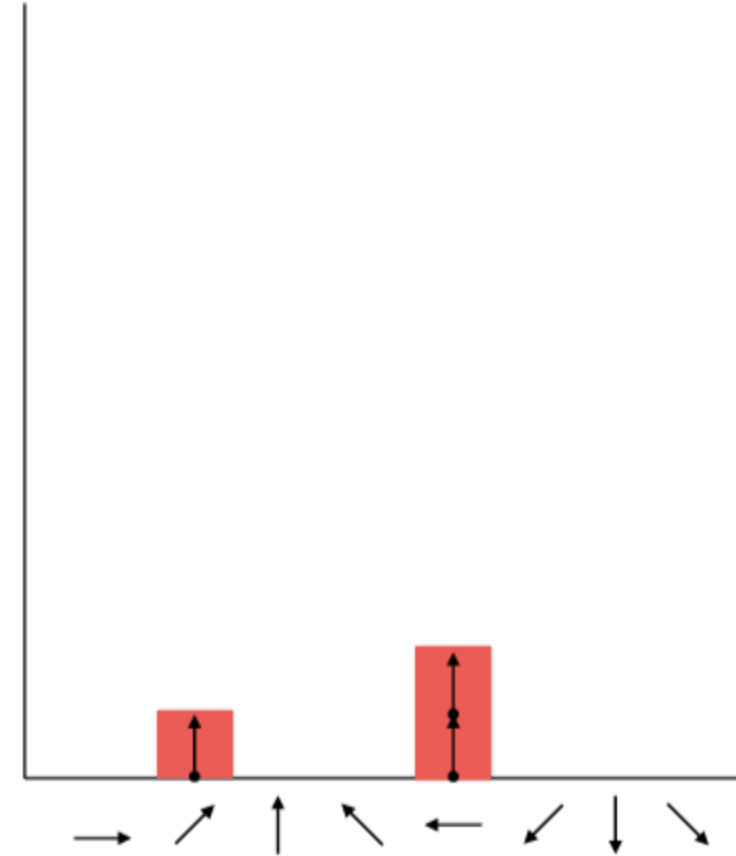


This visual example is taken from [Leonid Sigal slides](#) (The University of British Columbia)

SIFT orientation assignment

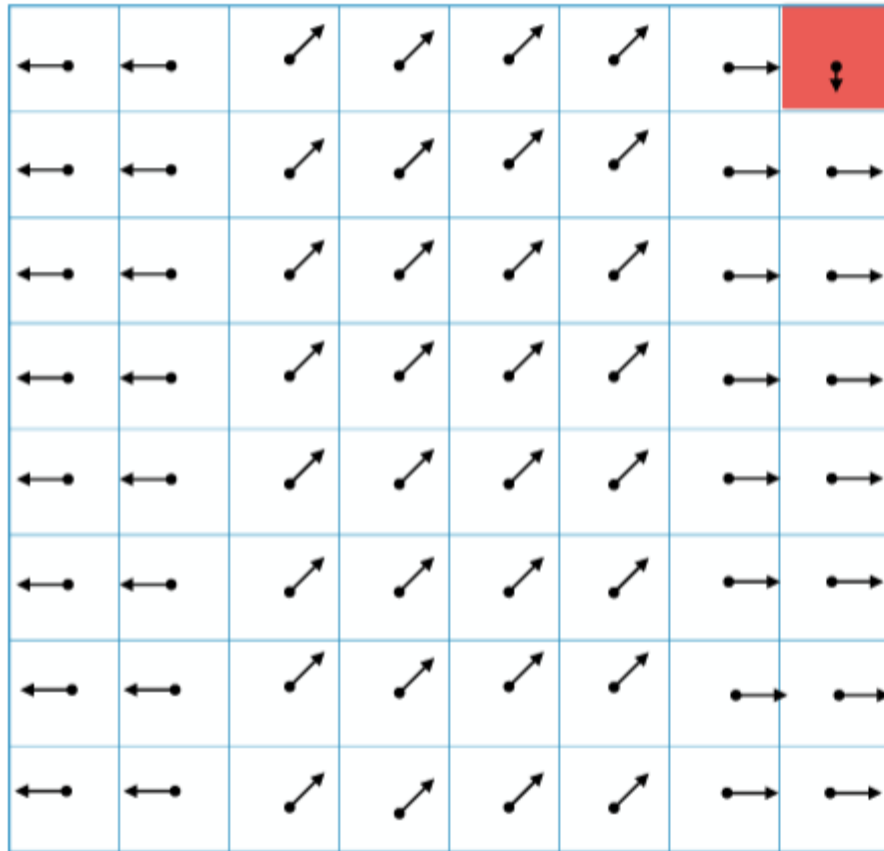


Arrows illustrate **gradient orientation** (direction)
and **gradient magnitude** (arrow length)

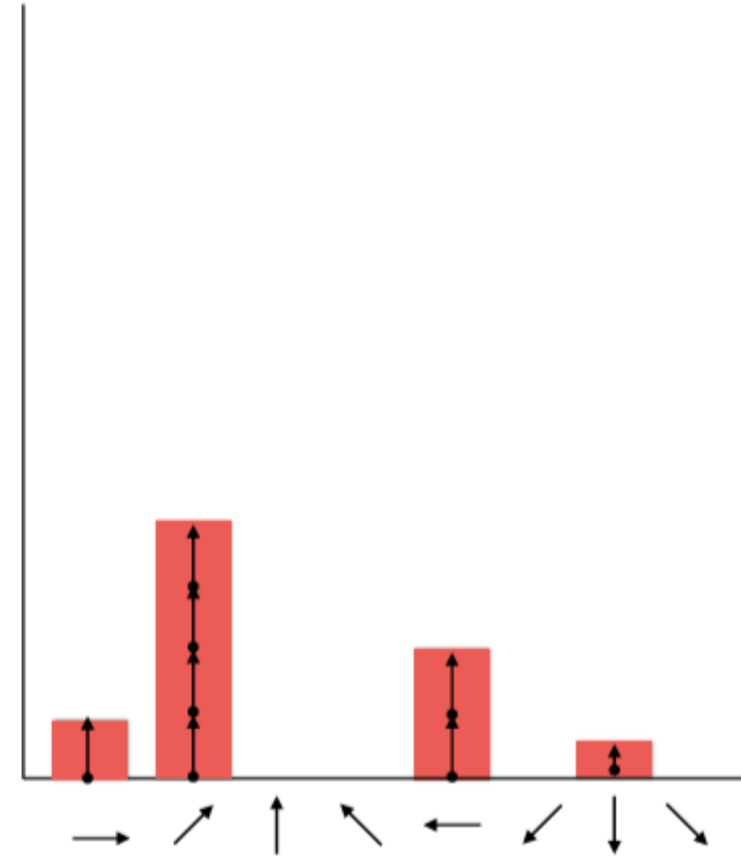


This visual example is taken from [Leonid Sigal slides](#) (The University of British Columbia)

SIFT orientation assignment

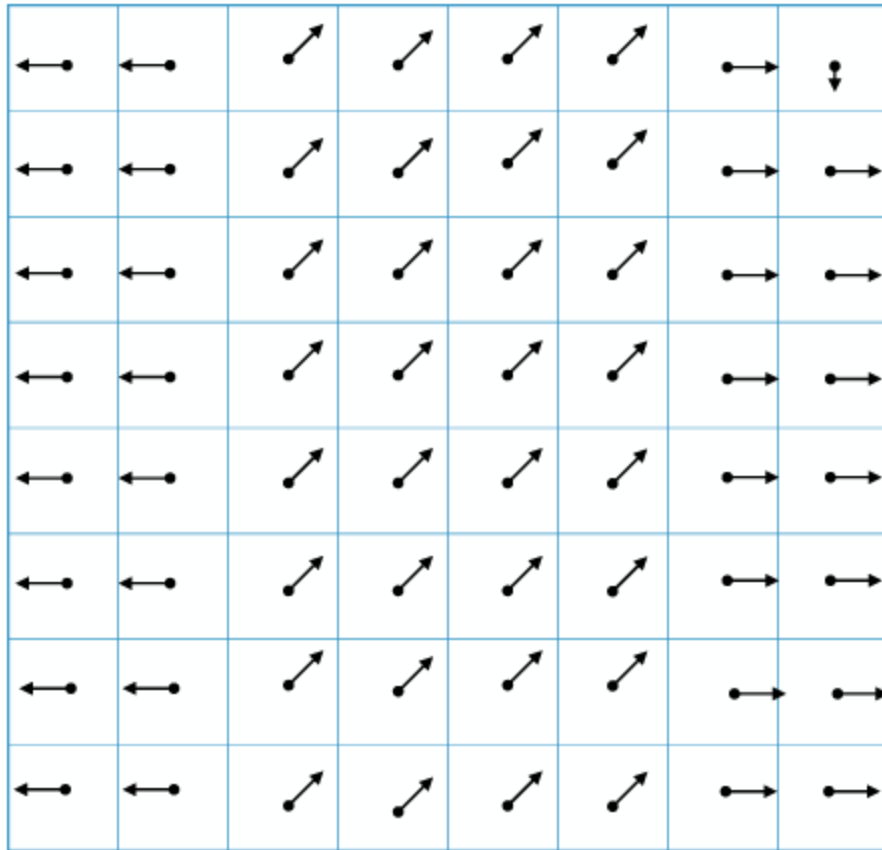


Arrows illustrate **gradient orientation** (direction)
and **gradient magnitude** (arrow length)

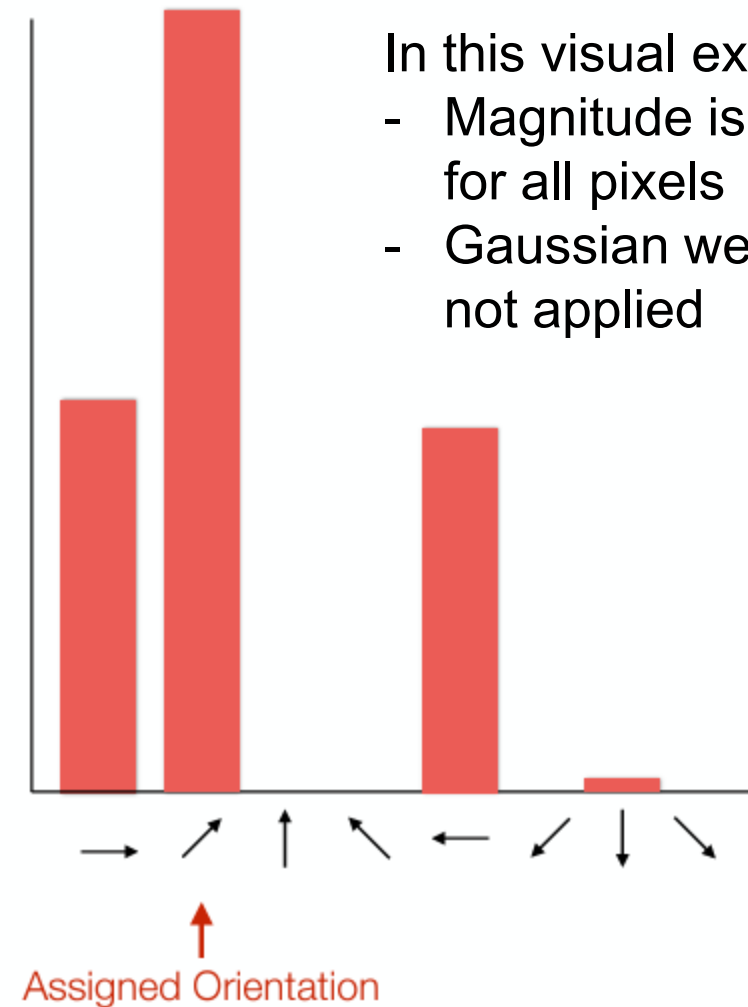


This visual example is taken from [Leonid Sigal slides](#) (The University of British Columbia)

SIFT orientation assignment



Arrows illustrate **gradient orientation** (direction)
and **gradient magnitude** (arrow length)



This visual example is taken from [Leonid Sigal slides](#) (The University of British Columbia)

SIFT descriptor

For each (x, y, scale, orientation), create descriptor:

- Using a **normalized patch after undoing the effect of rotation and scaling** (orientation and scale invariance)
- Divide a 16x16 window around the keypoint location into a 4x4 grid of cells in scale-space
- Compute a weighted orientation histogram for each cell (8 bins) using gradient magnitude
- Final descriptor: 16 cells * 8 orientations = **128 dimensional descriptor**
- Descriptor is normalized to unit length ($\vec{v}/\|\vec{v}\|$), then clipped (if >0.2 is saturated to 0.2), and finally renormalized to unit length to reduce the effects of illumination change.

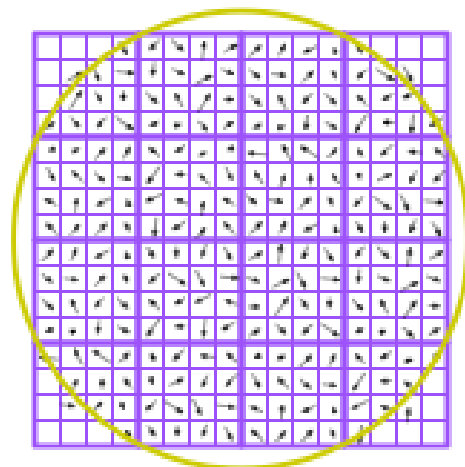
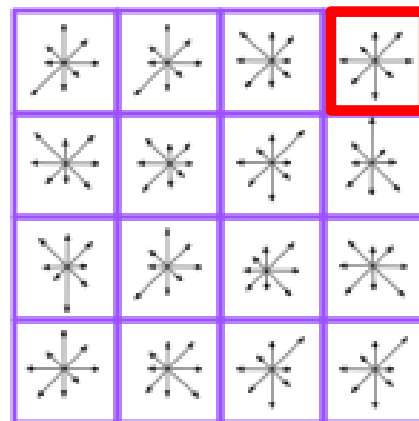
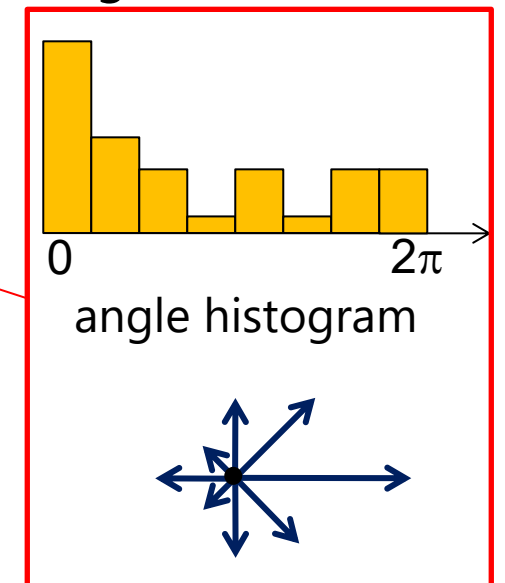


Image gradients



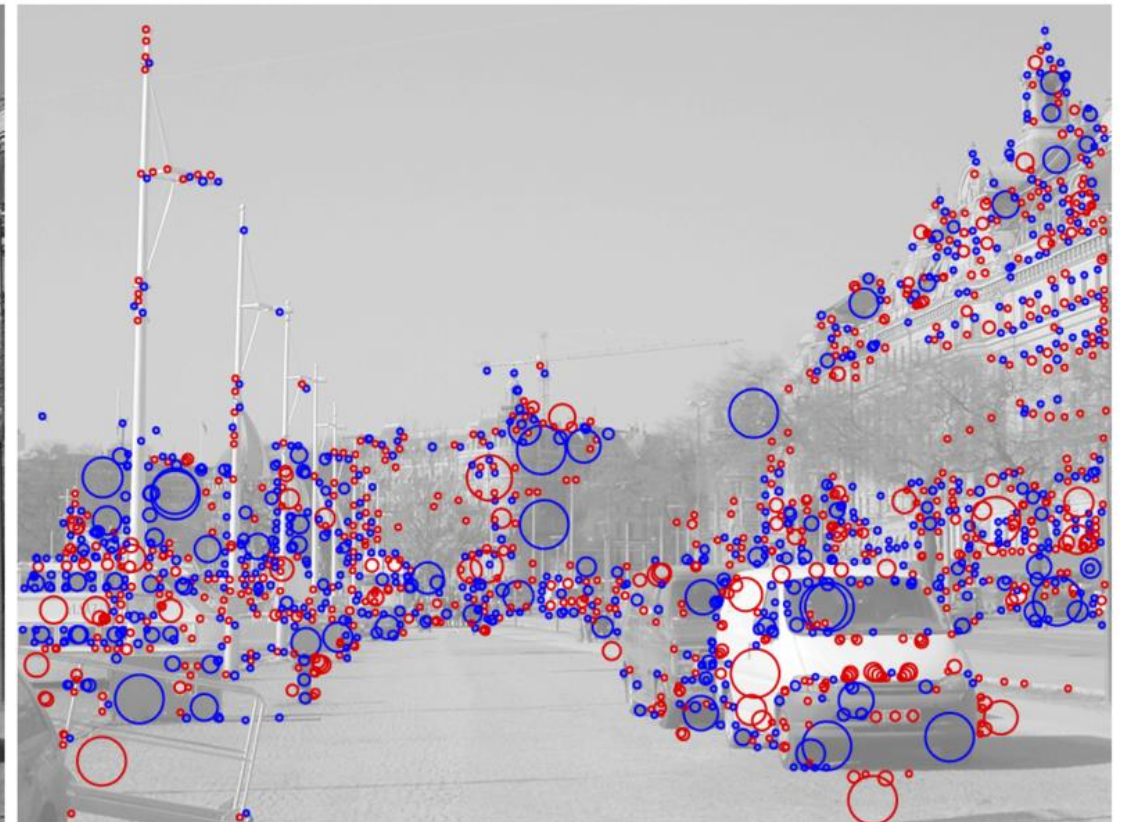
Keypoint descriptor



Properties of SIFT

“The SIFT descriptor is **invariant to translations, rotations and scaling transformations** in the image domain and **robust to moderate perspective transformations and illumination variations.** ”

Tony Lindeberg, http://www.scholarpedia.org/article/Scale_Invariant_Feature_Transform



Properties of SIFT

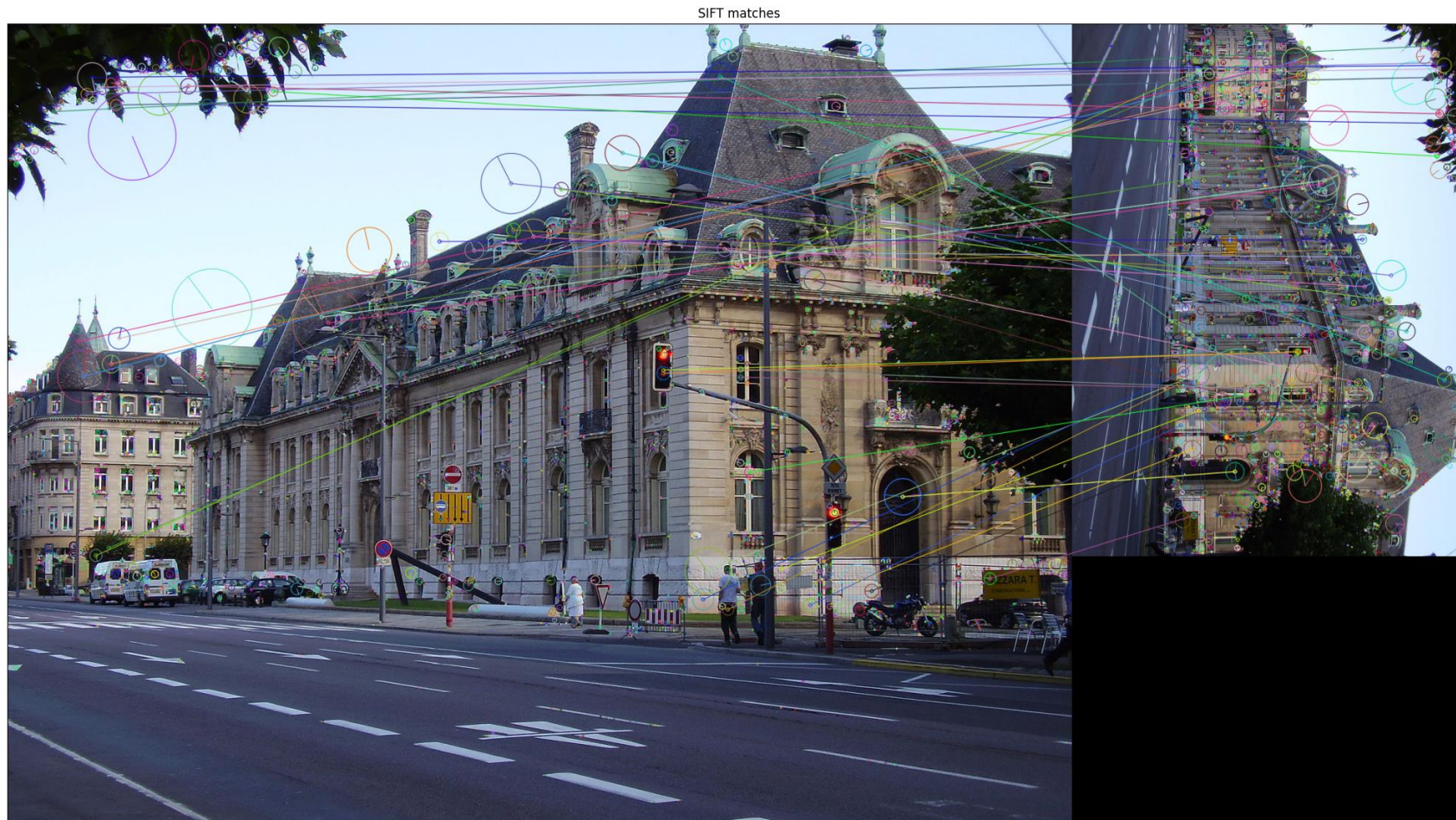
Extraordinarily robust matching technique

- Can handle in some cases relatively large changes in viewpoint (up to about 30-60 degrees, depending on the authors)
- Can handle some changes in illumination (sometimes even day vs. night (below))
- Pretty fast (hard to make real-time, but can run in $<1s$ for moderate image sizes)
- Lots of code available



Properties of SIFT

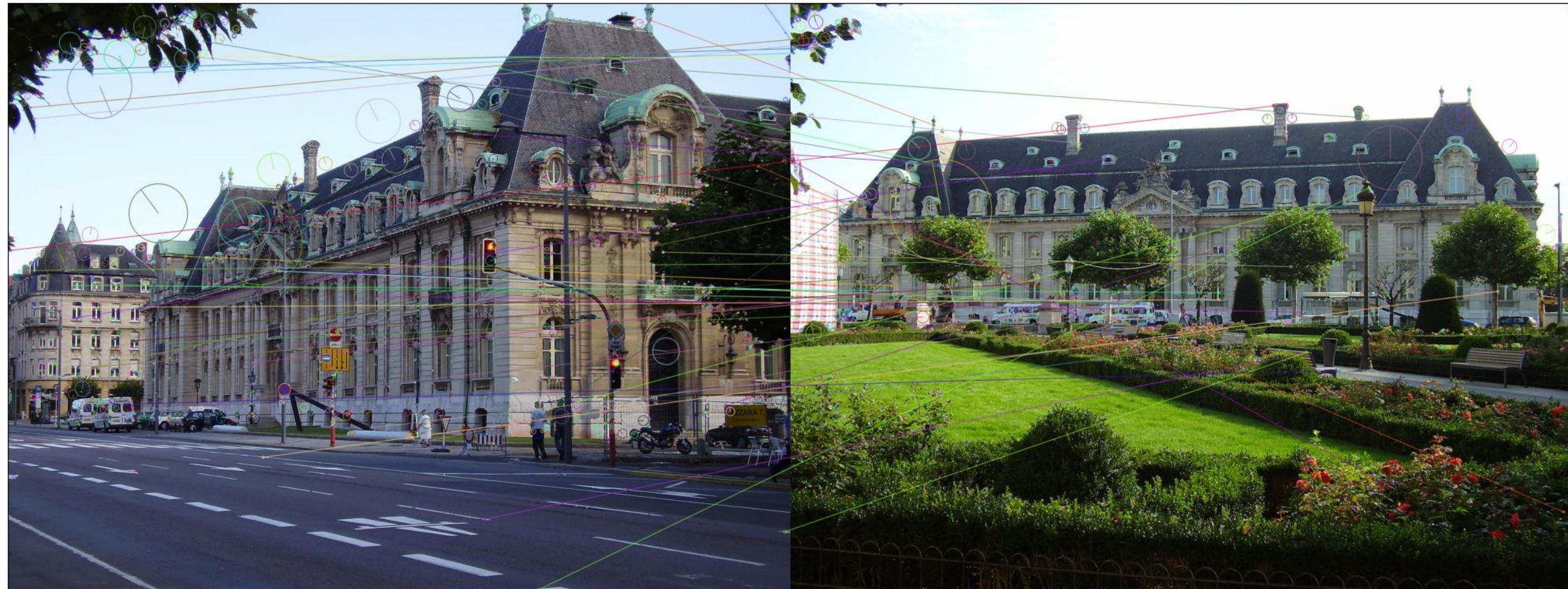
Invariant to translation, rotation and scale, and robust to clutter and changes in brightness. In the example below, we applied SIFT to two images (one is the scaled and 90° rotated version of the other one). We just display 50 matches.



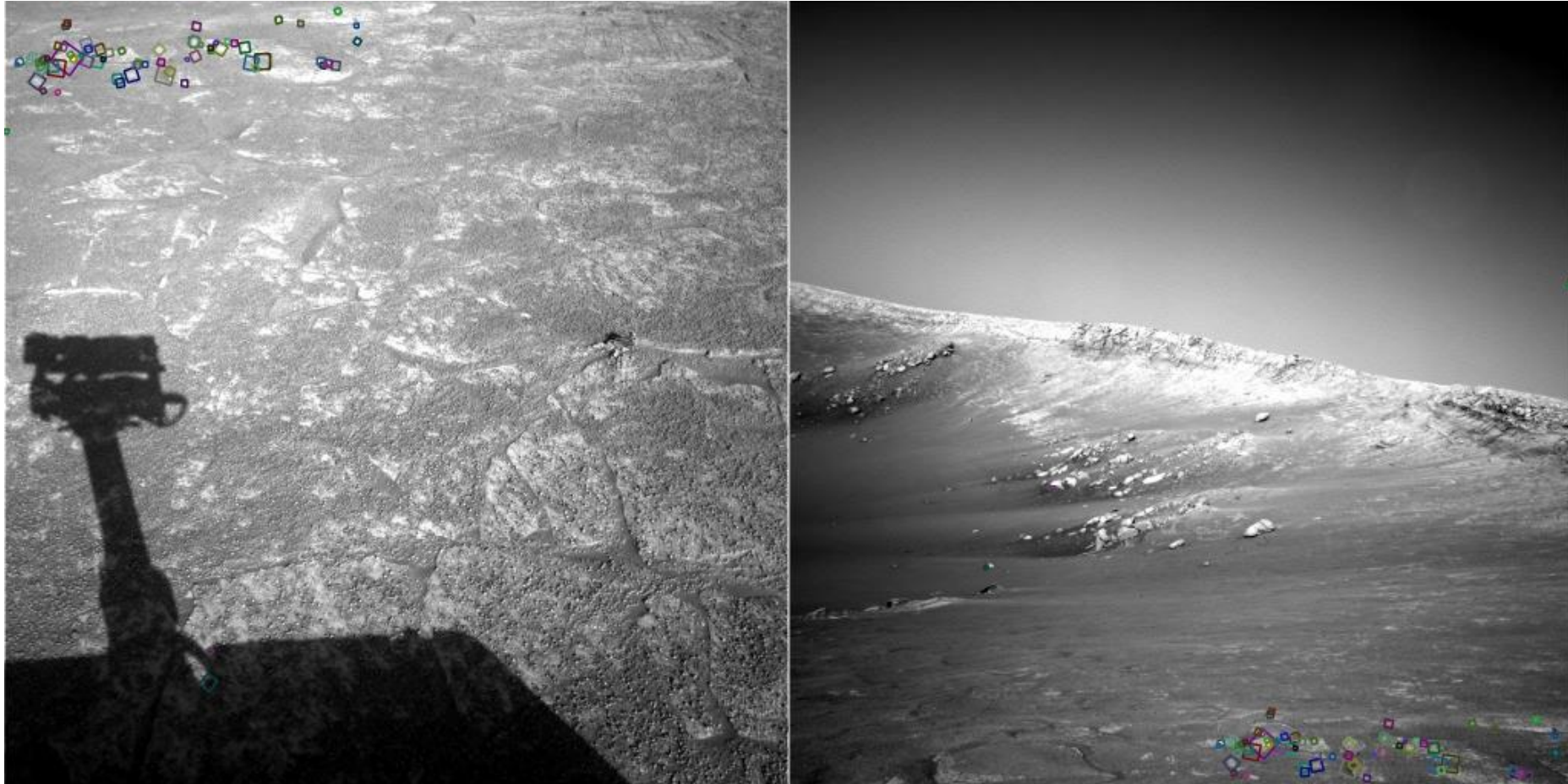
Properties of SIFT

If viewpoint differs a lot we could have problems...

SIFT matches



Very stable in difficult scenes



NASA Mars Rover images
with SIFT feature matches

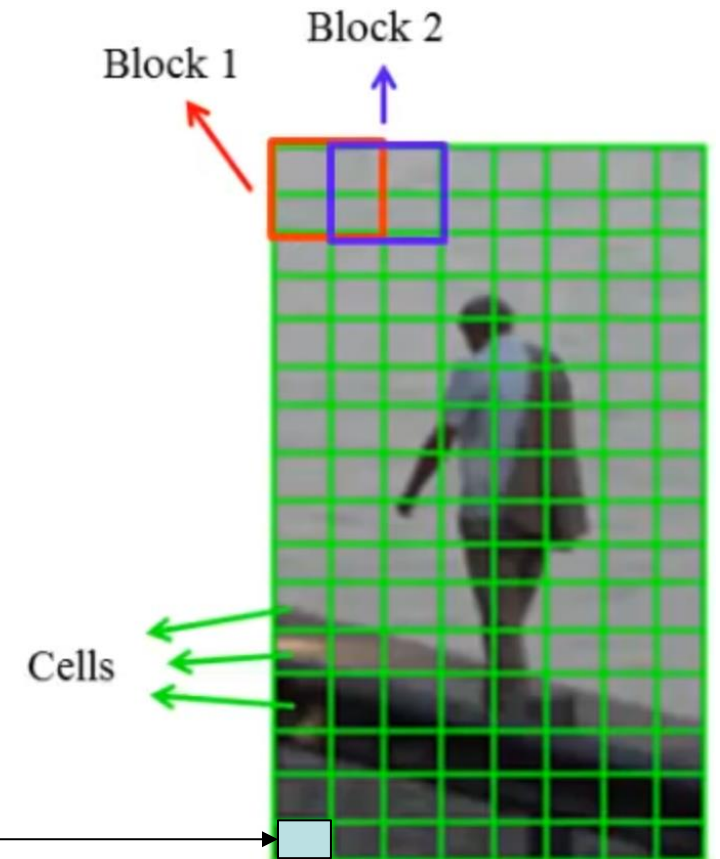
Histogram of Oriented Gradients (HOG)

HOG (Histogram of Oriented Gradients)

- Dalal, N., & Triggs, B. (2005). Histograms of oriented gradients for human detection. In *IEEE computer society conference on computer vision and pattern recognition (CVPR'05)* (Vol. 1, pp. 886-893). (>49.000 citations)
- Similar to SIFT, but it is computed on a dense grid of uniformly spaced cells → this is a global descriptor!
- The original work was focused on pedestrian detection.

HOG Feature Extraction

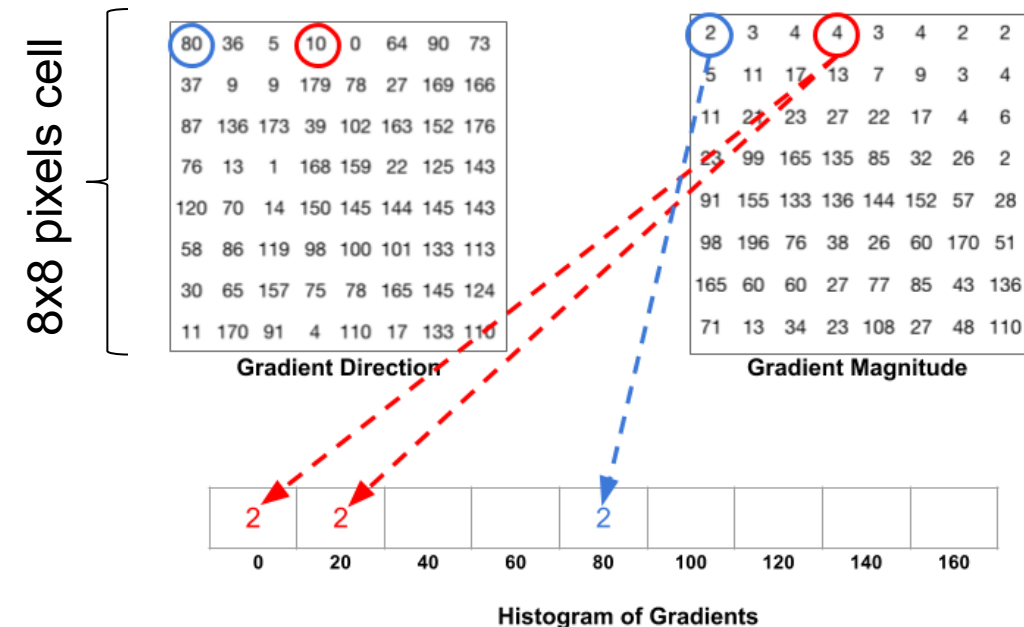
- 1) Compute gradients of the input image (64x128 in the original paper)
- 2) Create 16x16 pixels overlapping blocks
 - $7 \times 15 = 105$ blocks in total
 - Each block should consist of 2×2 cells with size 8x8 (each cell)



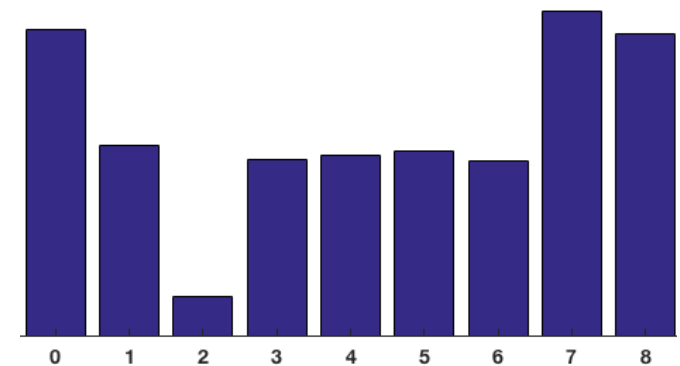
HOG Feature Extraction

3) Quantize the gradient orientation (unsigned gradients) of each 8x8 pixels cell into a 9 bins/orientations histogram

- The vote is the gradient magnitude.
- Interpolate votes linearly between neighboring bin centers.



Since 10 degrees is halfway between 0° and 20° , the vote by the pixel splits evenly into the two bins.



Example taken from <https://learnopencv.com/histogram-of-oriented-gradients/>

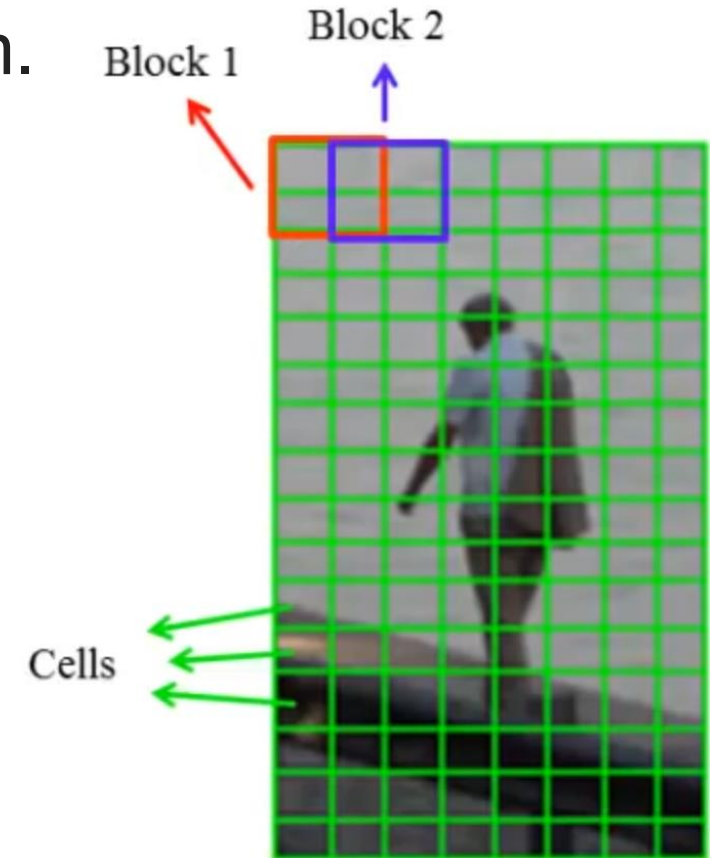
HOG Feature Extraction

3) Quantize the gradient orientation (unsigned gradients) of each 8x8 pixels cell into a 9 bins/orientations histogram

- The vote is the gradient magnitude.
- Interpolate votes linearly between neighboring bin centers.
- The vote can also be weighted with Gaussian to downweight the pixels near the edges of the block.
 - To reduce impact of window translations (with pixels in the edges moving to the adjacent blocks).
 - What happens in the center of the block has a larger impact.

HOG Feature Extraction

- 4) Normalize histograms on each block (9 bins x 4 cells/block $\rightarrow$ 36-d vector) to reduce the effects of illumination/lighting variation in the gradient magnitude
 - We divide each vector element by its L2 norm.
- 5) Concatenate histograms (we have one per cell)
 - 105 blocks x 4 cells x 9 orientations
= 3.780 dimensional feature vector



Take-home message

Key characteristics of robust feature descriptors:

- Local patterns (defining the feature's neighborhood)
- Gradient orientations (robust to changes in illumination)
- Histograms (often weighted, prioritizing detail at the center over the boundary, typically using a Gaussian window)

Other feature detectors/descriptors

Some Feature Detectors/Descriptors

- SURF: Speeded up robust features
 - Bay et al. (ECCV'06) (>25.000 citations)
 - 6 times faster than SIFT
 - Equivalent quality for object identification
- ORB
 - Rublee et al. (ICCV'11) (>16.000 citations)
 - FAST detector + BRIEF descriptor
 - Fast alternative to SIFT
- FAST
 - Rosten & Drummond (ECCV'06) (>7.500 citations)
 - Computationally efficient corner detector
- BRISK
 - Leutenegger et al. (ICCV'11) (>5.000 citations)
 - Binary descriptor
 - An order of magnitude faster than SURF in some cases
- FREAK: Fast Retina Keypoint
 - Alahi et al. (CVPR'12) (>2.800 citations)
 - Perceptually motivated
 - Can run in real-time
- KAZE features
 - Alcantarilla et al. (ECCV'12) (>2.000 citations)
 - Nonlinear scale space by means of nonlinear diffusion filtering
- AKAZE features
 - Alcantarilla et al. (BMVC'13) (>1.700 citations)
 - Fast KAZE features
- LIFT: Learned Invariant Feature Transform
 - Yi et al. (ECCV'16) (>1.750 citations)
 - Learned via deep learning
- BRIEF
 - Calonder et al. (TPAMI'11) (>1.000 citations)
 - Very fast binary descriptor.
 - Comparable performance to SURF and SIFT

Some Feature Detectors/Descriptors

Feature	Detection	Description	OpenCV	Published
Harris	Yes	No	Yes	1988
KLT	Yes	No	Yes	1994
LBP	No	Yes	Yes	1994
SIFT	Yes	Yes	Yes	IJCV 2004
FAST	Yes	No	Yes	ECCV 2006
SURF	Yes	Yes	Yes	CVIU 2008
BRIEF	No	Yes	Yes	ECCV 2010
ORB	Yes	Yes	Yes	ICCV 2011
BRISK	Yes	Yes	Yes	ICCV 2011
FREAK	Yes	Yes	Yes	CVPR 2012
SuperPoint	Yes	Yes	No	CVPR 2018
R2D2	Yes	Yes	No	NeurIPS 2019
DISK	Yes	Yes	No	NeurIPS 2020

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Data Science and Computational Intelligence